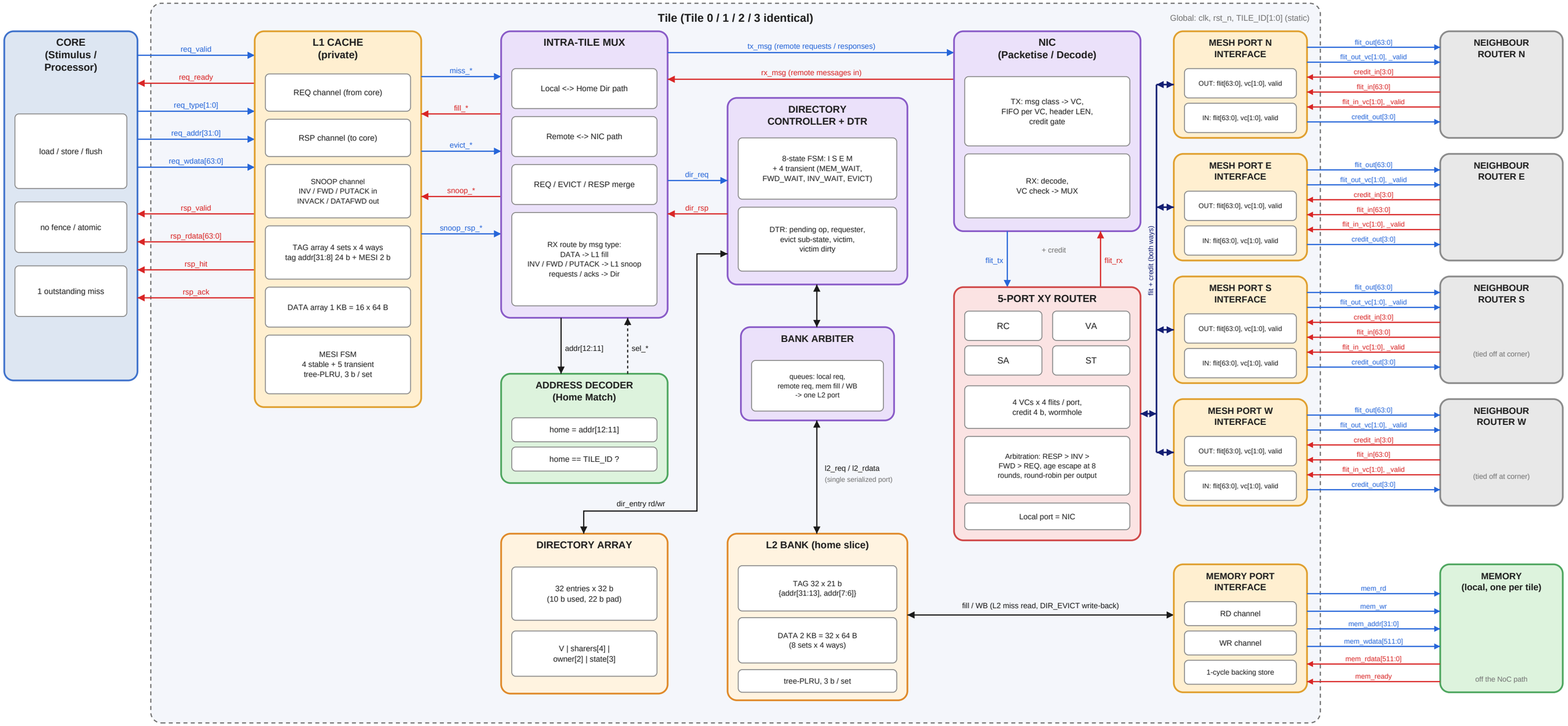
Tile Datapath (Cache-Coherent Tiled Multi-core) - Low Level Design (LLD)

One tile shown (Tile 0 port usage); all four tiles of the 2x2 mesh use the identical module. 64-bit flits. One local memory port per tile.



Legend

- Request path (core req, miss, tx, mem write)
- Response path (rsp, fill, rx, mem read)
- Control / address signal
- Internal control (decoder select)
- Mesh link: flit + credit (both ways)
- Interface block
- Logic / control block
- Decoder
- Router
- Storage (arrays, L2, memory)

Signal names are illustrative; replace with RTL names.

Key Signals (per Interface)

Interface	Signals
Core - L1	req_valid, req_ready, req_type[1:0] (load / store / flush), req_addr[31:0], req_wdata[63:0]; rsp_valid, rsp_rdata[63:0], rsp_hit, rsp_ack
L1 - MUX	miss_valid/ready/type (GETS, GETM)/addr[31:0]; evict_valid/ready/type (PUTS clean S/E, PUTM dirty)/addr[31:0]/data[511:0]; fill_valid/ready, fill_data[511:0], fill_excl, UNBLOCK; snoop_valid/ready/type (INV, FWD_GETS, FWD_GETM, PUTACK)/addr[31:0]/src[1:0]; snoop_rsp_valid/ready/type (INVACK, DATAFWD)/addr[31:0]/data[511:0]
MUX - Dir	dir_req_valid/ready/msg[3:0]/src[1:0]/addr[31:0]/data[511:0] (local + remote requests and acks); dir_rsp_valid/ready/msg[3:0]/dst[1:0]/addr[31:0]/data[511:0]
MUX - NIC	tx_valid/ready, tx_msg[3:0], tx_dst[1:0], tx_addr[31:0], tx_data[511:0]; rx_valid/ready, rx_msg[3:0], rx_src[1:0], rx_addr[31:0], rx_data[511:0]
NIC - Router	flit[63:0], flit_valid, vc[1:0] (set at source, never rewritten), credit[3:0] (each direction)
Router - Neighbour	per port: flit_out/in[63:0], flit_out/in_vc[1:0], flit_out/in_valid, credit_in/out[3:0]
Dir - L2	l2_req_valid/ready/rw/addr[31:0], l2_wdata[511:0], l2_rdata[511:0], l2_hit (through the bank arbiter, one access at a time)
L2 - Memory	mem_rd, mem_wr, mem_addr[31:0], mem_wdata[511:0], mem_rdata[511:0], mem_ready
VCs / packets	VC0 REQ (GETS, GETM, PUTS, PUTM); VC1 FWD (FWD_GETS, FWD_GETM); VC2 RESP (DATA, DATAFWD, INVACK, UNBLOCK, PUTACK); VC3 INV (INV). Control packet = 1 flit. Data packet (PUTM, DATA, DATAFWD) = 9 flits (1 header + 8 x 64 b); length carried in the header LEN field.

Address Map (32-bit address)

Bits	Field	Use
[31:13]	tag (19 b)	L2 tag (with [7:6])
[12:11]	home (2 b)	Home tile: 00=T0 01=T1 10=T2 11=T3*
[10:8]	L2 set (3 b)	8 sets in the L2 bank
[7:6]	L1 set (2 b)	4 sets in the L1
[5:0]	offset (6 b)	Byte in 64 B line

Stored tags: L1 = addr[31:8] (24 b); L2 = {addr[31:13], addr[7:6]} (21 b). Home bits are constant per bank and are not stored.

L1 set [7:6] and L2 set [10:8] are disjoint, so every L1 set can reach all 8 L2 sets.

home == TILE_ID: request stays in this tile (MUX -> Dir), with no NIC or NoC traversal.

home != TILE_ID: request goes MUX -> NIC -> Router -> mesh (XY).

Each tile's memory port serves only lines homed here.

* TILE_ID[1] = X, TILE_ID[0] = Y; every 2x2 tile is a corner, so exactly two mesh ports are active (T0: N, E | T1: S, E | T2: N, W | T3: S, W). Numbering is proposed; match your RTL.

Working Flow (Summary)

1. Core issues load/store/flush to L1. A hit completes locally.
2. L1 miss (GETS/GETM) or eviction (PUTS/PUTM) goes to the MUX with address and type.
3. Decoder compares addr[12:11] with TILE_ID to find the home tile.
4. Local home: MUX sends it directly to the Dir Controller. Remote home: MUX sends it to the NIC.
5. NIC packetises (header LEN), picks the VC by message class, waits for a credit.
6. Router (RC, VA, SA, ST) forwards the flits by XY routing to the home tile. The VC is never rewritten.
7. Home Dir reads the entry and opens a transaction (DTR); it sends FWD/INV if needed, or accesses the L2 bank through the bank arbiter.
8. The L2 bank is read first; on a bank miss the local memory port is read and the line fills the L2 (DIR_EVICT first if a victim is needed; dirty victims write back).
9. DATA returns via NIC and MUX; L1 installs the line (E/S/M by the exclusive bit); the core completes.
10. L1 sends UNBLOCK (VC2) to release the directory transient. Evictions are released by PUTACK.
11. Remote INV/FWD reach the L1 on the snoop channel, serviced independently of an outstanding miss; INVACK/DATAFWD go back to the home Dir.
12. Limits: one miss per L1, one transaction per directory bank.